

## Folder 04: Data Extraction & Synthesis

### 1. Content Overview

This folder provides the structured extraction tables developed from the 110 primary studies included in this SLR. A hybrid AI-assisted and human-verified extraction workflow was used to support technical depth, consistency, and accuracy across the four research questions (RQs):

- **RQ1 (Dataset Evaluation and Applicability):** Identifies the main characteristics of datasets used in ML-based insider threat detection, including dataset name/version, source type, log modality, size, duration, users, labels, attack scenarios, and availability where reported.
- **RQ2 (Dataset Evaluation & Applicability):** Analyzes the strengths, limitations, and applicability of different dataset types, including synthetic, real-world, multi-source, and domain-specific datasets.
- **RQ3 (Data Preprocessing Techniques):** Examines the methods used to transform raw data and security logs into machine-learning-ready representations, including cleaning, aggregation, session/window construction, encoding, normalization, imputation, outlier handling, dimensionality reduction, class-imbalance handling, feature construction, and train-test splitting.
- **RQ4 (Data Related Challenges):** Evaluates data-centric challenges that affect reliability, reproducibility, and generalizability in insider threat detection research, including synthetic-data dependence, limited real-world access, class imbalance, weak ground truth, privacy/anonymization, dataset heterogeneity, inconsistent preprocessing reporting, and cross-dataset generalization.

### 2. Extraction Methodology

To improve transparency and consistency, the following extraction pipeline was used:

1. **AI-Assisted Synthesis:** NotebookLM and Google Gemini were used to support the initial extraction of technical information from the full-text PDFs of the 110 included studies.
2. **Manual Verification:** The extracted information was checked against the original source papers by the authors to confirm accuracy and relevance.
3. **Cross-Checking:** Key technical details, such as dataset type, data sources, preprocessing steps, window sizes, normalization methods, and reported limitations, were manually reviewed to reduce AI-generated errors and ensure scientific accuracy.

### 3. Document Descriptions

| File Name                                              | Description                                                                                                                                                                    |
|--------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 04_01_Master_Data_Extraction_Table_Revised_RQ1_RQ4.Xls | <b>Master extraction table</b> Master extraction table containing structured data for all 110 included studies. Each row represents one study, with columns mapped directly to |

|                              |                                                                                                                                                                                                                   |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                              | dataset characteristics (RQ1), dataset advantages and limitations (RQ2), preprocessing techniques (RQ3), and data-related challenges (RQ4).                                                                       |
| 04_02_RQ1_Dataset_Matrix.pdf | <b>Focused Dataset Report:</b> Focused dataset matrix primarily supporting RQ1, with related notes for RQ2. It summarizes dataset names, characteristics, source types, availability, strengths, and limitations. |

#### 4. Data Validation Note

All extracted information was reviewed for consistency by two reviewers. The qualitative synthesis reported in Section V of the manuscript was derived from the structured extraction data contained in these files.